UNIVERSITAT POLITÈCNICA DE CATALUNYA BARCELONATECH

FIB

PROP Project - Hidato

Team 12.4

version 3.0

Aghzaf El Hachimi, Mohamed	(mohamed.aghzaf)
Cifuentes Salamí, Félix	(felix.cifuentes)
Miquel Mitjans, Rafel	(rafel.miquel)
Ollers Rigo, Joan Pere	(joan.pere.ollers)

Table of contents

1. The application....2

2. Execution....2

3. Login and Register....3

4. Menu view....4

5. Functionalities....6

5.1. Hidatos generation....6

5.2. Hidatos resolution and validation....8

5.3. Interactive game....10

5.4. Hidatos creation and validation....12

5.5. Hidato repositories....14

5.6. Time control....15

5.7. Ranking....16

5.8. Save and load matches....18

5.9. Difficulty and hints....19

1. The application

Our application is a 2D puzzle game called Hidato. The goal is to complete a path of consecutive numbers on a grid, from 1 to the maximum cell value, ensuring all adjacent numbers connect. This guide provides a complete walkthrough of the game's features. If you haven't heard of Hidato, take a look at the example below!

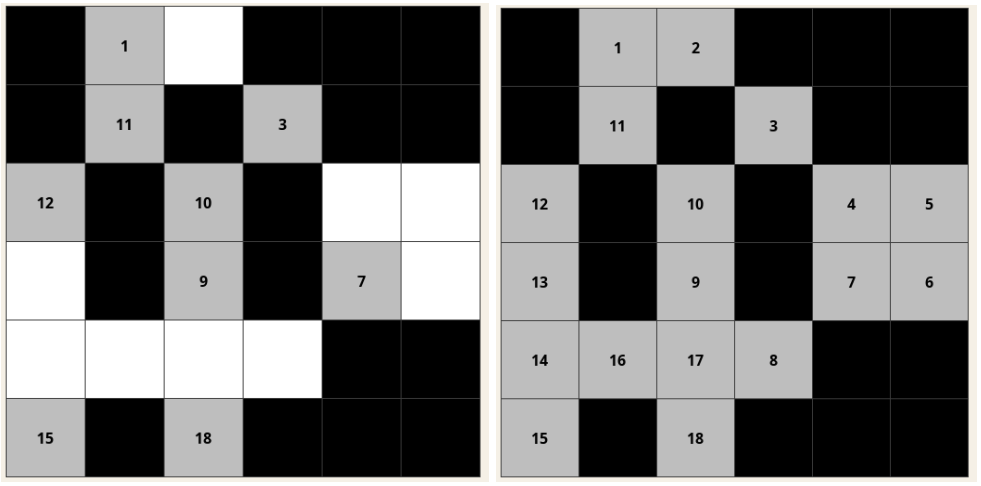


Figure 1. Example of an incomplete hidato and its solution.

2. Execution

Prerequisites: Java JDK 21.x and Linux/MacOS terminal (Shell, Bash, Zsh…).

Installation, compilation and execution:

1. Download the project (.tar.gz)
2. Insert the following command in the terminal

```
Shell
mkdir ~/hidatogame && tar -xzf ~/Downloads/<filename>.tar.gz -C ~/hidatogame
```

3. Once extracted, compile and execute using the following command

```
Shell
cd ~/hidatogame/FONTS && ./gradlew build && ./gradlew run
```

A window will pop up with the login page of the game.

3. Login and Register

The first thing that will appear when running the app is the login prompt. In case it's your first time entering the game, you'll have to register. Click the "Register" button as seen in *Figure 2* and create your account inserting a username and a password, which you'll have to confirm, as seen in *Figure 3*. Then, log in and enter the game.

It is worth mentioning that all errors you will encounter are in the form of pop-up windows with a message indicating what caused it. All of them can be closed and you will be able to continue or restart what you were doing.

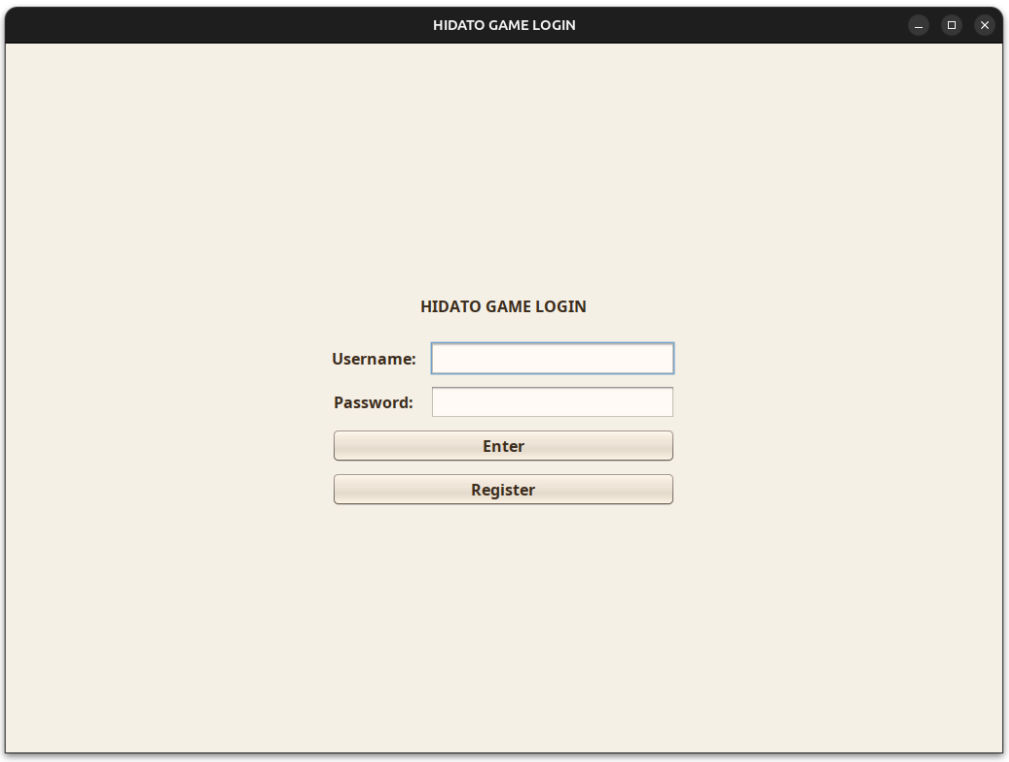


Figure 2. Login view.

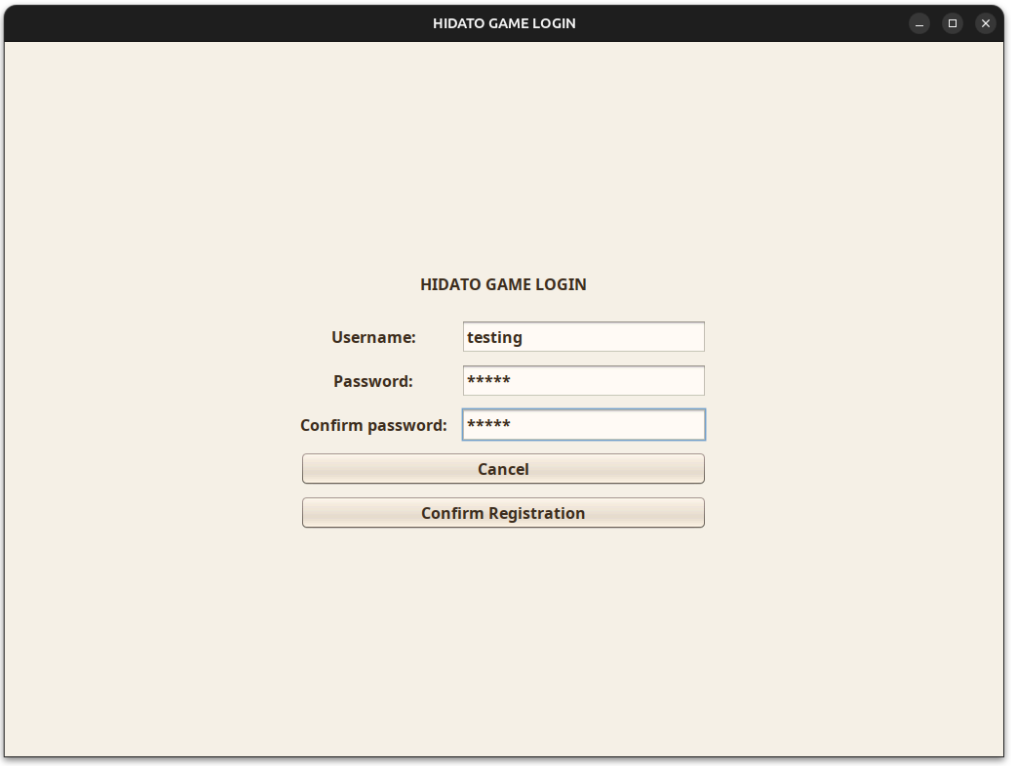


Figure 3. Register view

4. Menu view

Once you log in, the principal menu will appear, with 6 possible playable options and a "Logout" button. Before getting to know what each option allows you to do, bear in mind that what we understand as a "match" is simply the hidato currently being played. Therefore, those options are:

1. **Play Casual**

   You can start a match by loading an existing hidato, resuming an existing match or simply generating a new random match choosing the difficulty.

2. **Play Ranked**

   If you are competitive, this is your mode. You will be able to start a random ranked match choosing the desired difficulty, and if you manage to solve it, a score will be obtained and added directly to the global ranking.

3. **Generate Hidato**

   You will be asked to generate a random hidato based on your constraints; the topology or cell type, which can be in the shape of squares, triangles or hexagons, the adjacency type, which will determine how the path goes and can only be sides or sides and vertices, the difficulty and the number of cells in the path (cells being a hole or a wall do not count here).

4. **Create/Import Hidato**

   This mode allows total freedom to create a hidato. You basically have to present it either selecting a CSV file or writing the hidato in directly in a text box with the appropriate format.

5. **View Ranking**

   Screen with the top 100 best user scores. The score is only obtained from correctly solving completely a hidato from a ranked match.

6. **Consult Lists**

   Its like a mini-filesystem where you'll manage hidatos. You'll see the repositories or lists of hidatos and the hidatos the selected list has. You'll be able to Create/Delete repositories and Add/Delete hidatos from them, except from the "All" list which is read-only and accessible to all users.

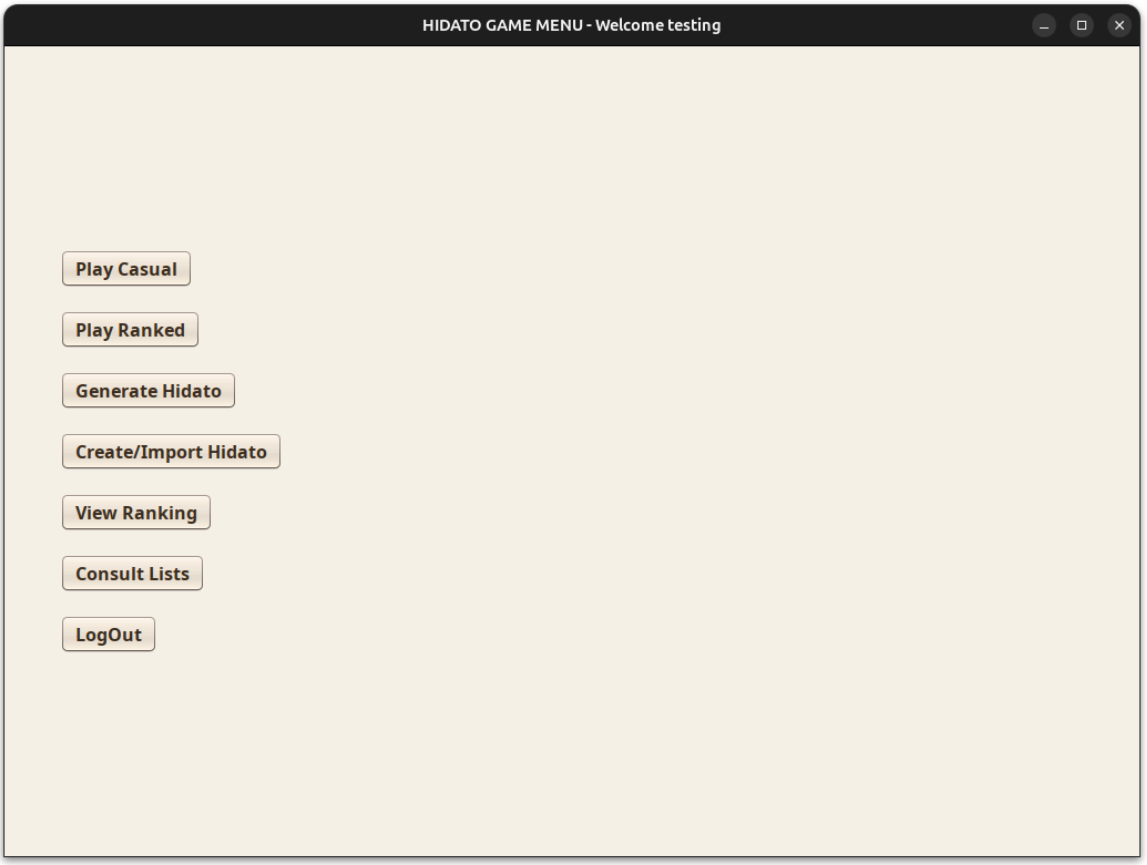


Figure 4. Menu view.

5. Functionalities

5.1. Hidatos generation

From the menu, select "Generate Hidato" option. You will see a screen like *Figure 5*. There are three (3) drop-down lists, consisting of the topology, the adjacency and the difficulty chosen. We detailed what those options are in the previous section. Then, you must input the number of cells the path will have.

** If Hexagons topology is selected then you'll see the Adjacency list disappear as this shape has only the sides option viable.*

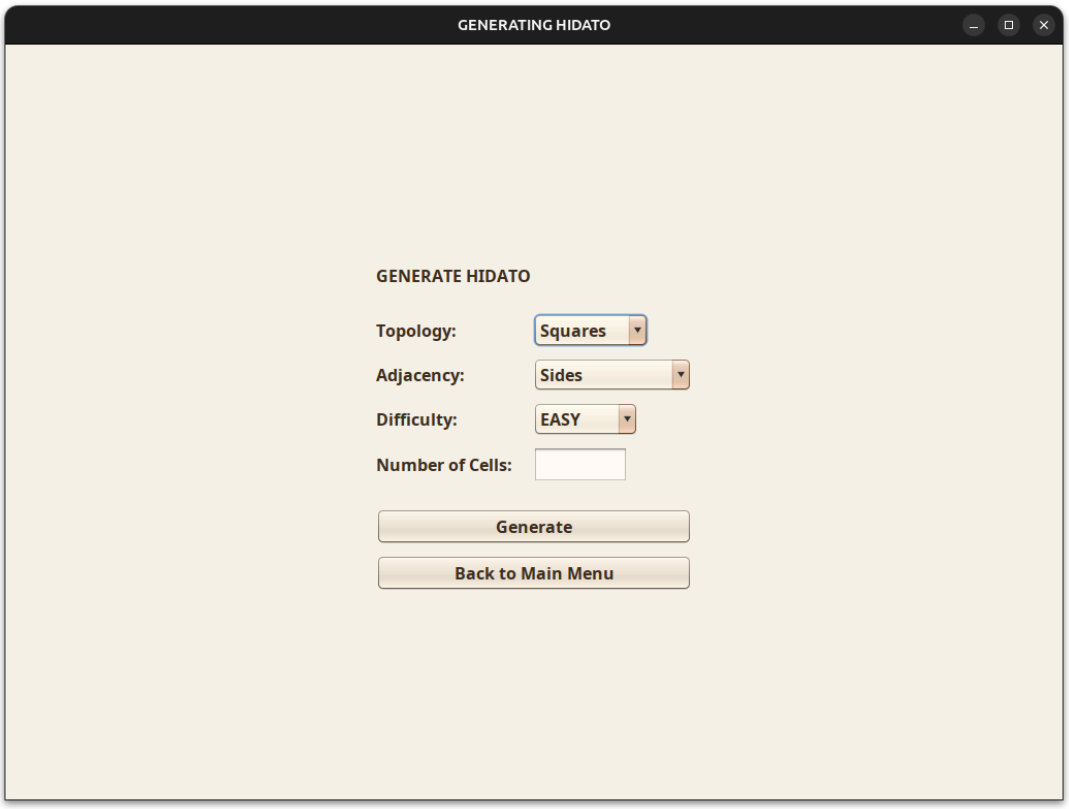


Figure 5. Generate Hidato view.

Let's generate a hidato as an example. Immagine we want a *Triangles* topology with *sides plus vertices* adjacency, a *MEDIUM* difficulty[1] and a 20 cell path. Look at *Figure* 6 to know what that looks like.

[1] *MEDIUM* difficulty means, independently of the cell number, that half of cells (50%) will be empty for you to solve. See more information on difficulty levels at section 5.9.

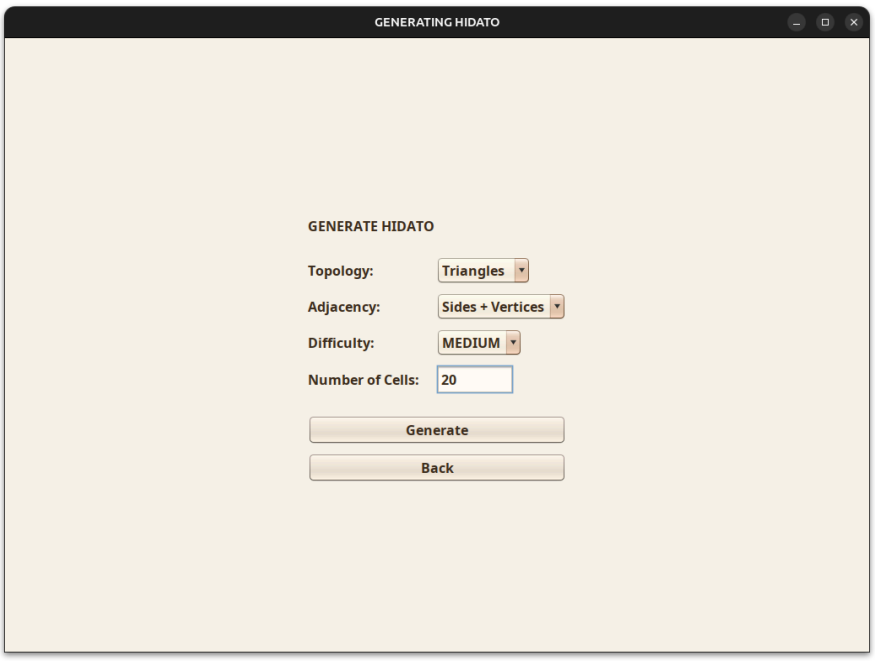


Figure 6. Generate Hidato view example.

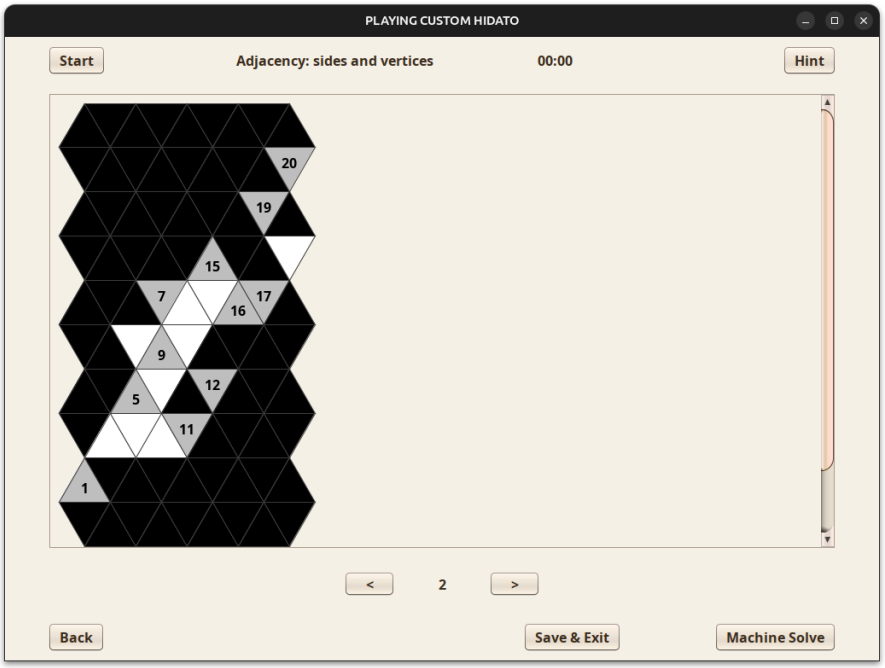


Figure 7. Generated hidato from the example constraints.

This mode only has one error possible, related with the inserted number of cells.

For a hidato to be valid it is required to have a minimum of 3 cells and a maximum of 200, both included. If you fail to do so, *Figure 8* shows the possible error messages.

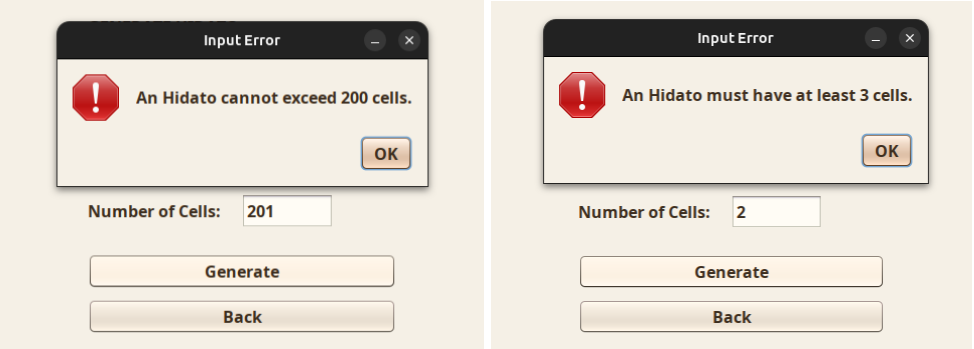


Figure 8. Error messages for Generate Hidato mode.

5.2. Hidatos resolution and validation

When a match is being played you will see the screen as in *Figure* 9 below. The only thing that changes each new match is, naturally, the hidato itself.

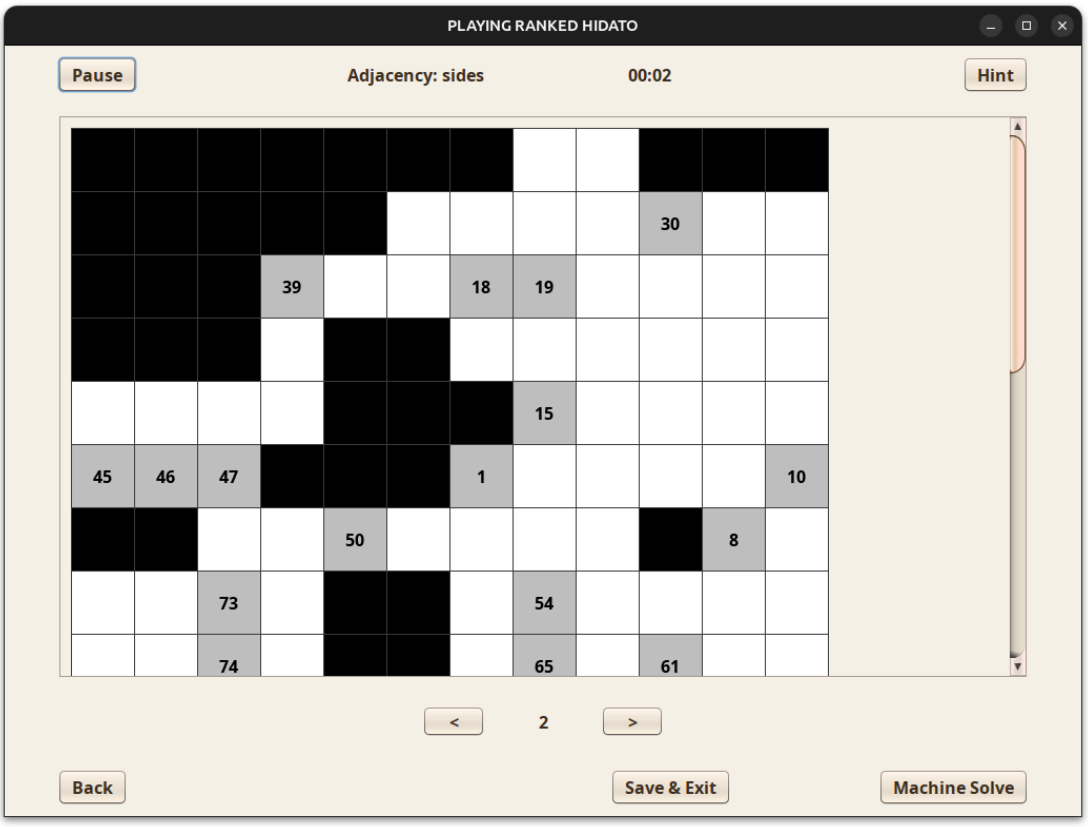


Figure 9. Hidato match example view.

What does each component do?

1. **Pause/Start button:** pauses or starts/restarts the chronometer. When a match is paused the hidato disappears to prevent users from cheating.
2. **Adjacency:** labels the hidato adjacency type.
3. **Chronometer:** computes the time elapsed and updates each second. Controlled by Start/Pause button and showed in a MM:SS format.
4. **Hint button:** gives you a random solved cell. There is no limit for you to ask hints, but each of them will affect negatively to your score.
5. **Number selectors buttons:** when you click on an empty cell of the hidato, the number it will get is based on the number between the two selectors "<" and ">". You can change to the previous or the next available number pressing each one of them, respectively.
6. **Save & Exit button:** your mom just called you to have dinner? Don't worry, press this button and save the match for when you are ready to play again[2].
7. **Machine Solve button:** validates if the hidato in the current match is solvable and if it is, it solves it and presents it to the user with the time the CPU used. Obviously, no points are awarded if you use this feature.
8. **Back button:** returns to the menu (*Figure 4*). All other screens with a *Back* button return you to the menu, to make navegability smooth and easy.
9. **Horizontal and vertical scrollers:** if the hidato is too big for the window display, scroll bars will appear to let you see all cells.

Let's suppose we are in the match showed in *Figure 9* above. The resolution functionality resides in the *Machine Solve* button explained above. Let's press on it! Now we got *Figure 10* solution and pop-up information.

[2] See section 5.8. on saving and loading matches for more information.

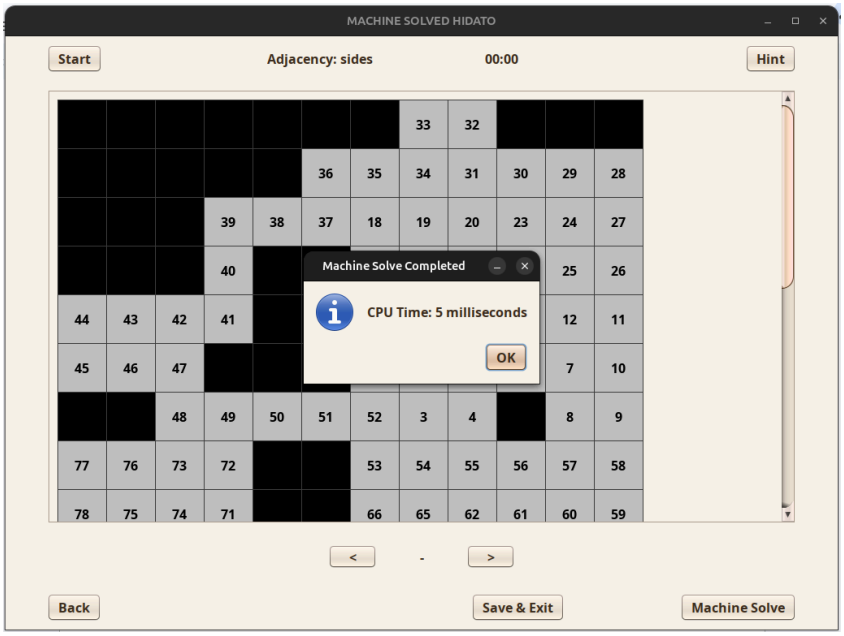


Figure 10. Hidato match example solved by the system and pop-up information.

5.3. Interactive game

To play a match and solve the hidato you will have to fill each empty cell at your own responsability. The cells you filled have a distinctive cyan colour, as shown in *Figures 11* and *12*. The cells obtained from a hint are also cyan, to help you track where the hint you got is, but those are not editable as they are 100% correct. If you click on a valid filled cell you will erase the number it had and make it empty. Each click on a cell updates the selected number, either to the next available or to the erased one.

Only when all cells from the path are filled you will get a pop-up notification on whether the proposed solution is correct (*Figure 11*, where also a score is awarded as the match is Ranked) or incorrect *(Figure 12)*. If its correct, you may press *OK* to return to the Menu. If its incorrect, you may press *OK* and continue playing, as the timer is still on!

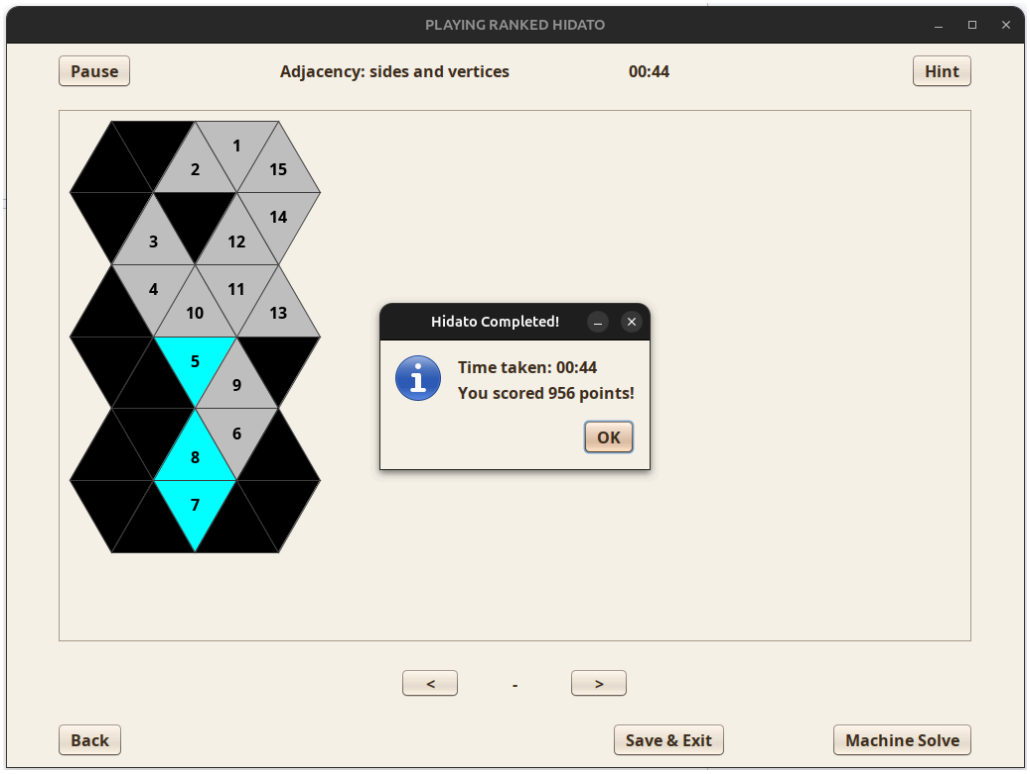


Figure 11. Interactive game correct solution example.

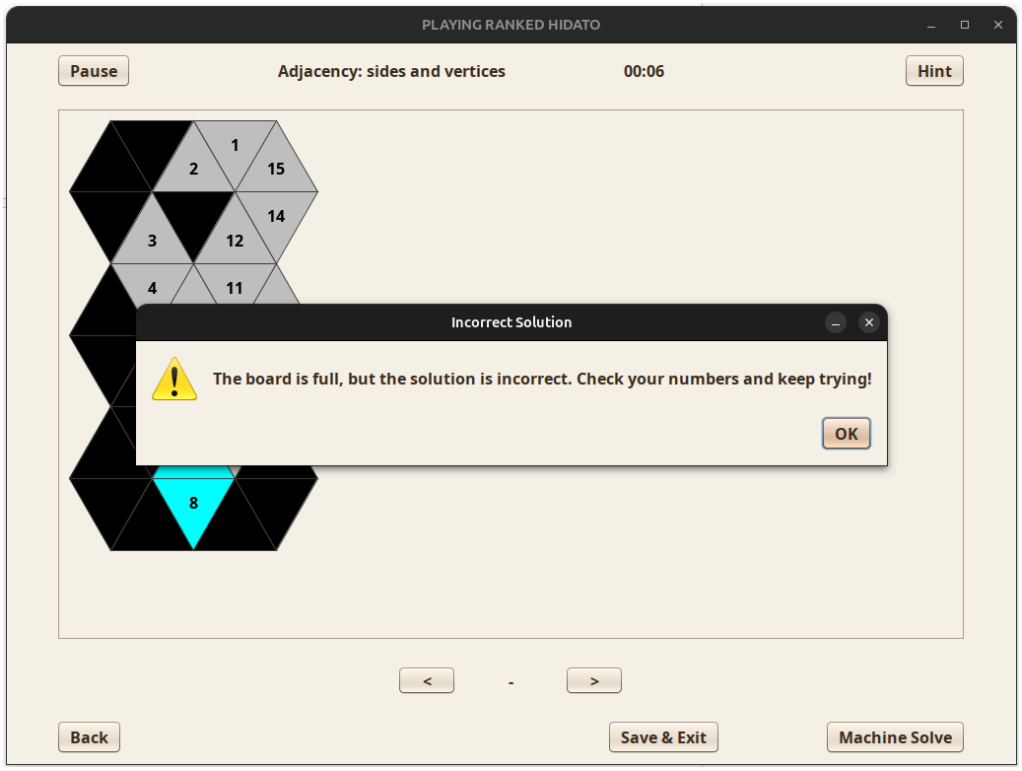


Figure 12. Interactive game incorrect solution example.

5.4. Hidatos creation and validation

From the menu, select *Create/Import Hidato* mode. You will see a screen like *Figure 13* below. Initially the big text area is empty, but you can paste a hidato in its CSV format or press the *Select File* button and access the /EXE folder where we have put some example hidatos for you to use.

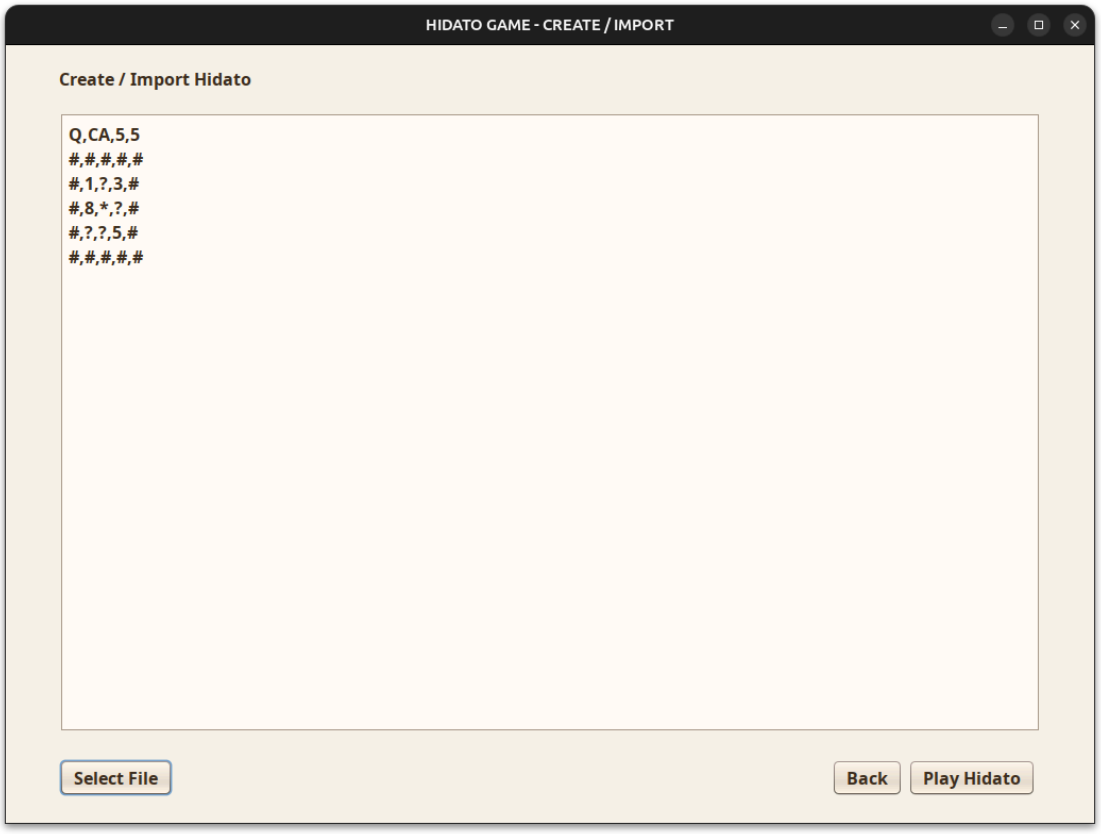


Figure 13. Create/Import Hidato view.

When you press *Play Hidato* button, the system will validate that the format is correct. Careful! That does not mean that the hidato is solvable, but that the format is valid to paint a hidato board. Error message is shown in *Figure 14*.

If the hidato format is valid, a match will start. As the hidato is imported or created by you, we do not guarantee that is solvable, so you may check it by trying *Machine Solve*. If it has no solution it will give the error message shown in *Figure 15*. Both errors

are recoverable by pressing *OK* or closing the pop-up and rewriting the hidato or going back to the menu.

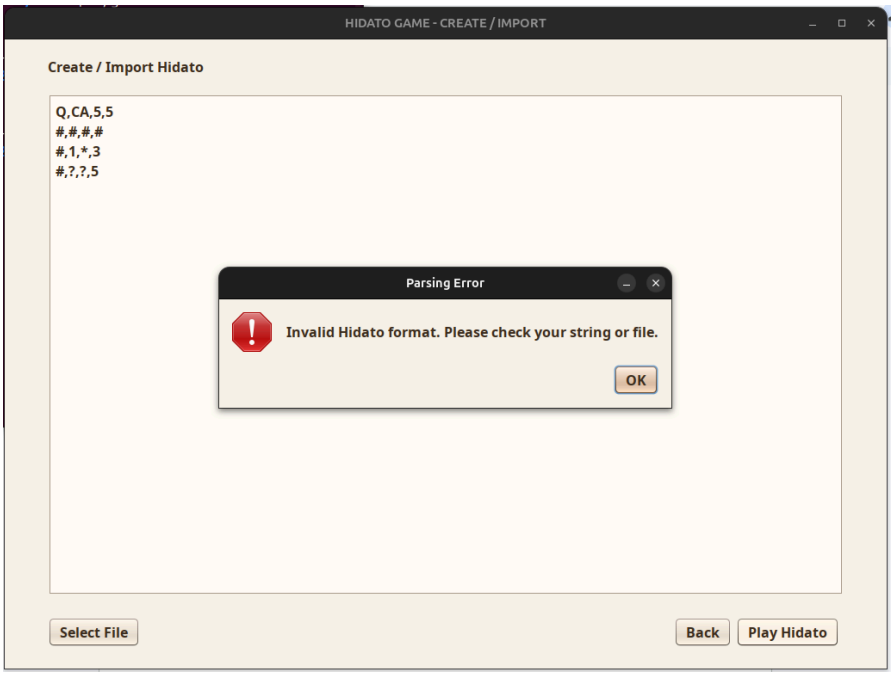


Figure 14. Create/Import Hidato format error. Expected 5x5 grid but provided a 3x4.

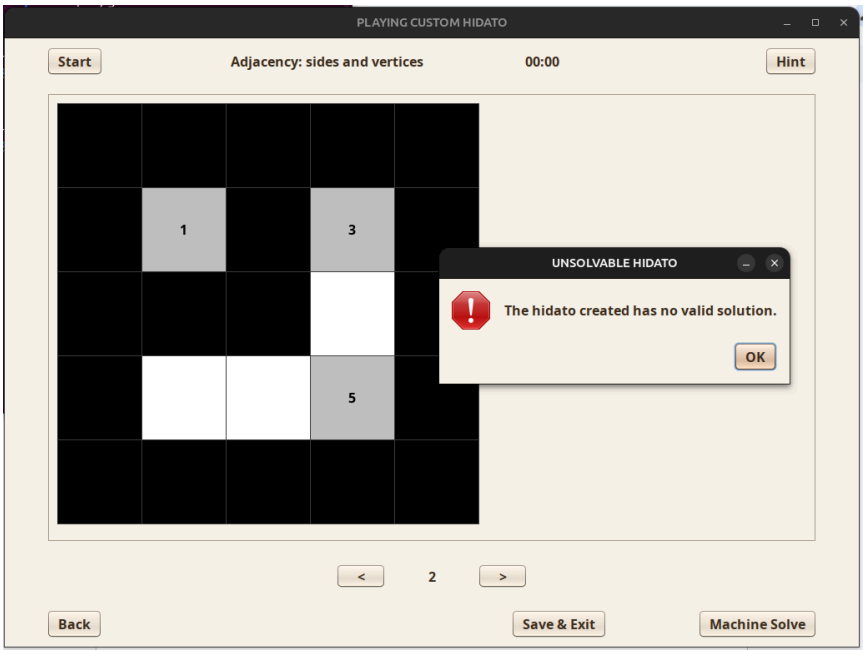


Figure 15. Create/Import Hidato match unsolvable error.

5.5. Hidato repositories

When a new hidato is created it is automatically saved in a repository called "all", accessible from all users. That occurs when a player starts solving a new casual match, a ranked match or generates a hidato. Where to view the repository? Accessing *Consult Lists* view from the menu. Not only you will see the "all" list with all the hidatos created until now, but also you'll be able to create or delete new lists and add or delete hidatos to those lists, as shown in *Figure 16*.

The list grows downwards, that is, the last hidato created is the last in the list. In the example it would be *Hexagons · 12x5 (29316ddf…)*. To help you identify your created hidatos they will appear with the information:

*Topology · Rows*Cols* (short identifier)

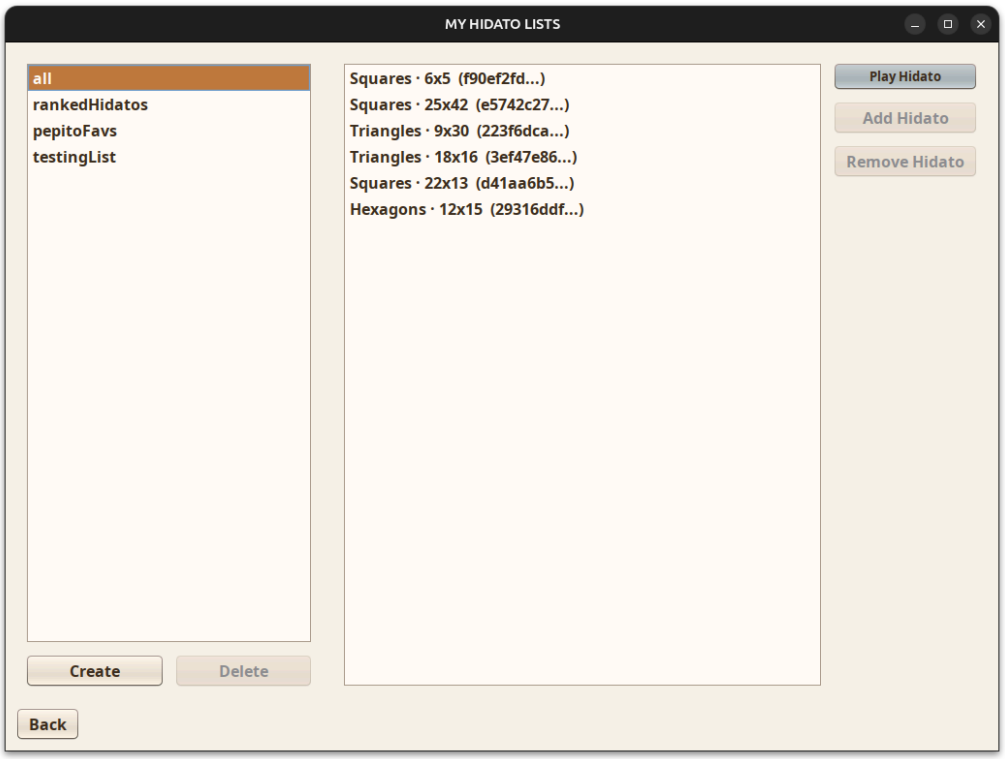


Figure 16. Consult Lists view.

You can also consult those lists from Play Casual menu, which is accessed pressing the *Play Casual* button inside the main menu. Like you can see in *Figure 17*, if you click on *Load Hidato* you'll be redirected to the view in *Figure 16*, therefore being able to select a hidato and play it directly pressing *Play Hidato* at the top right corner.

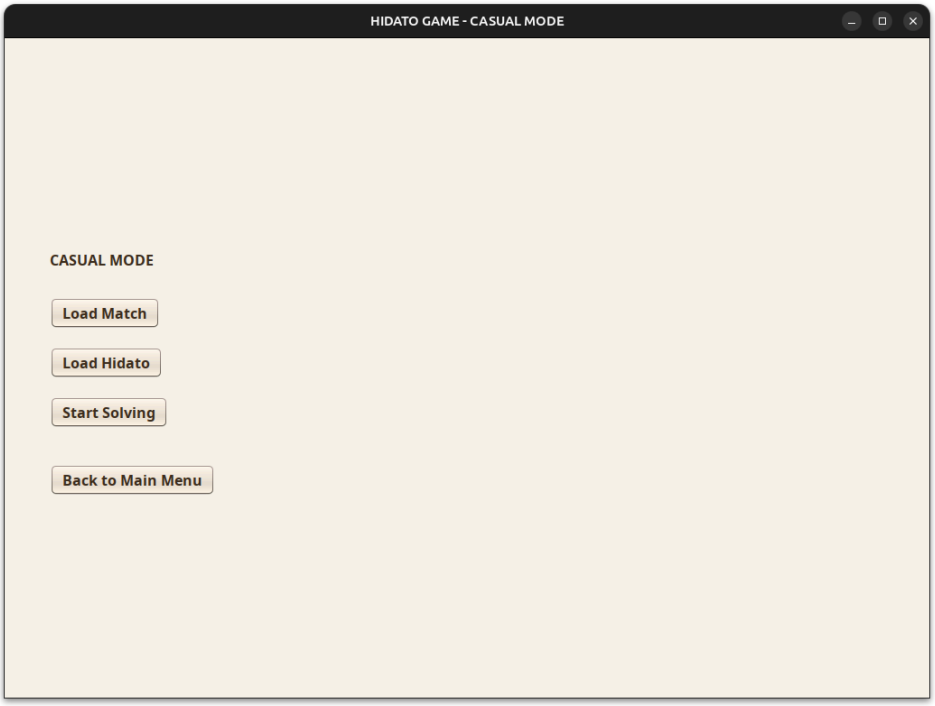


Figure 17. Play Casual view with Load Hidato option.

5.6. Time control

The match only starts when the chronometer starts ticking, either automatically or after a *Start* button press from the user. The matches started through A) automatically start, whereas the matches started through B) are in stand-by waiting the user to click *Start*.

A) *Load Match* and *Start Solving* from Play Casual and *Play Ranked*.

B) *Load Hidato* from Play Casual or Consult Lists, *Generate Hidato* and *Create/Import Hidato* modes.

5.7. Ranking

From the main menu, press *View Ranking* button. You will see a screen with all the users registered ordered by score. The list only shows the 100 best users. Each entry in the ranking consists of (1) #Position, (2) Username, (3) Total score obtained in ranked matches. Disclaimer! If you save an existing ranked match through *Save&Exit* button you lose the opportunity to get a score, but you may still continue the match loading it from *Play Casual* view. No errors in this option aswell.

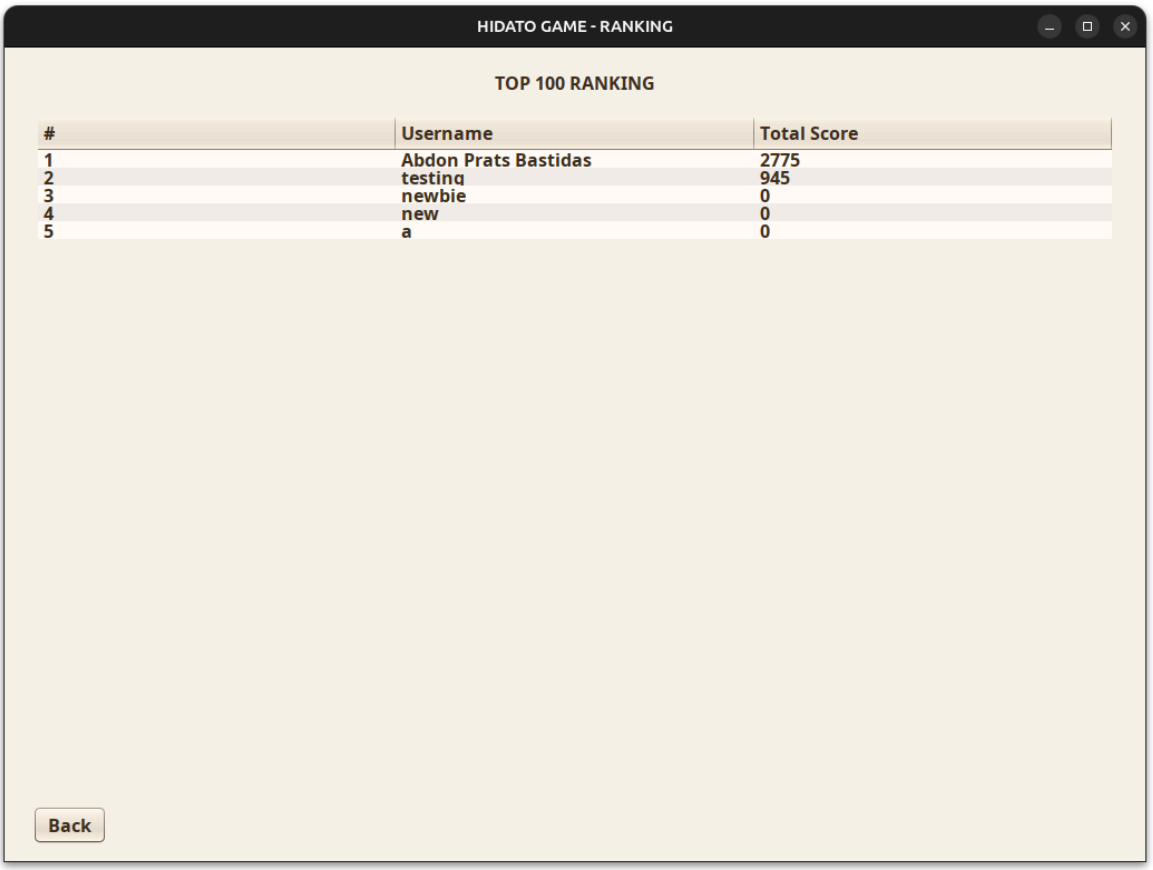


Figure 18. View Ranking example.

You may be asking: well, I can only obtain a score after solving correctly a hidato in a ranked match, but how is that score given?

The score system is implemented as it is shown in the *Table 1* below, but basically consists in starting with the perfect maximum score and getting penalties throughout the match.

Difficulty	Starting score	Hint penalty (per hint)	Time penalty (per second)
Easy	1000	100	1
Medium	2000	200	2
Hard	3000	300	3

Table 1. Scoring system

Therefore, let's say you solved a medium difficulty hidato in 01:15 and you performed so well that you didn't need any hints. Therefore the formula is:

$$Score = 2000_{base\ points} - 0_{hint\ penalty} - 75_{seconds} * 2_{time\ penalty} = 1850\ points$$



Figure 19. Example on computing a score.

5.8. Save and load matches

You have total freedom to *Save&Exit* a match whenever you want. Even a completed match! Therefore, you will always have stored the one that made you climb up the top 1 of the global ranking and all of its data.

In particular, the data saved is: current hidato board status, including your filled cells to let you modify them when you resume the game, current time elapsed and number of hints used. Other internal metadata is saved aswell.

All matches saved are visible clicking the *Load Match* button inside Play Casual view, which is again accessible through the menu. It opens a pop-up window with a drop-down list containing all saved matches. The last match saved is the last on the list, so scroll down until you find it. Each match has a unique identifier.

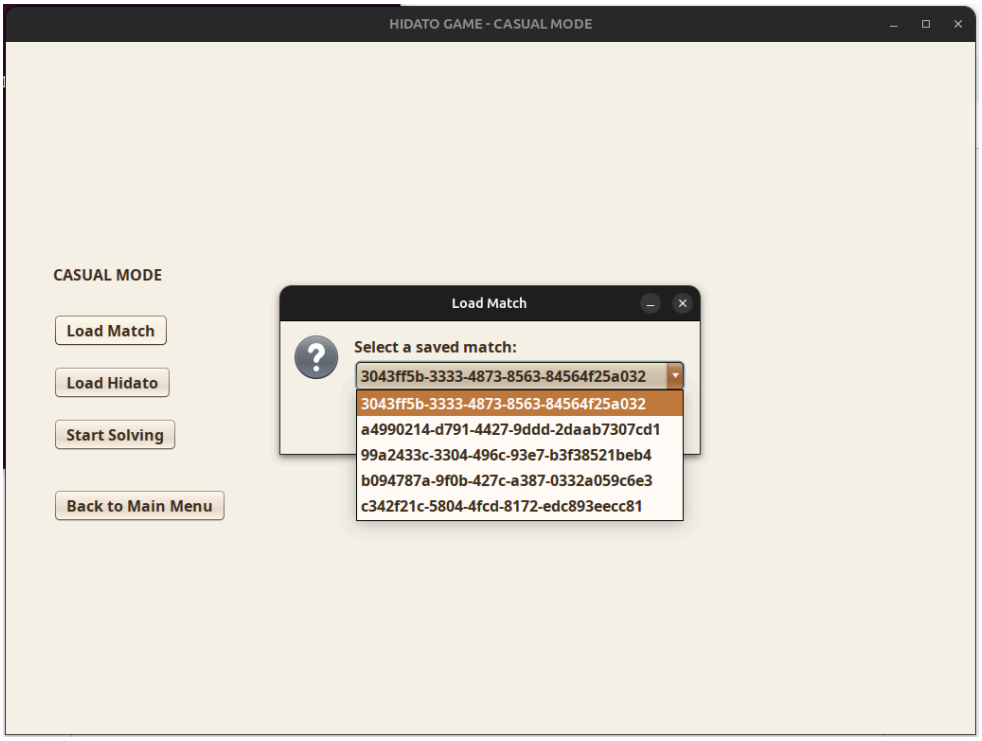


Figure 20. Load match pop-up window example.

5.9. Difficulty and hints

Difficulty levels are only important in Play Casual (*Start Solving* new random match), Play Ranked and Generate Hidato modes, as the user must select a level. It can be:

- EASY. The 25% of cells are empty.
- MEDIUM. The 50% of cells are empty.
- HARD. The 75% of cells are empty.

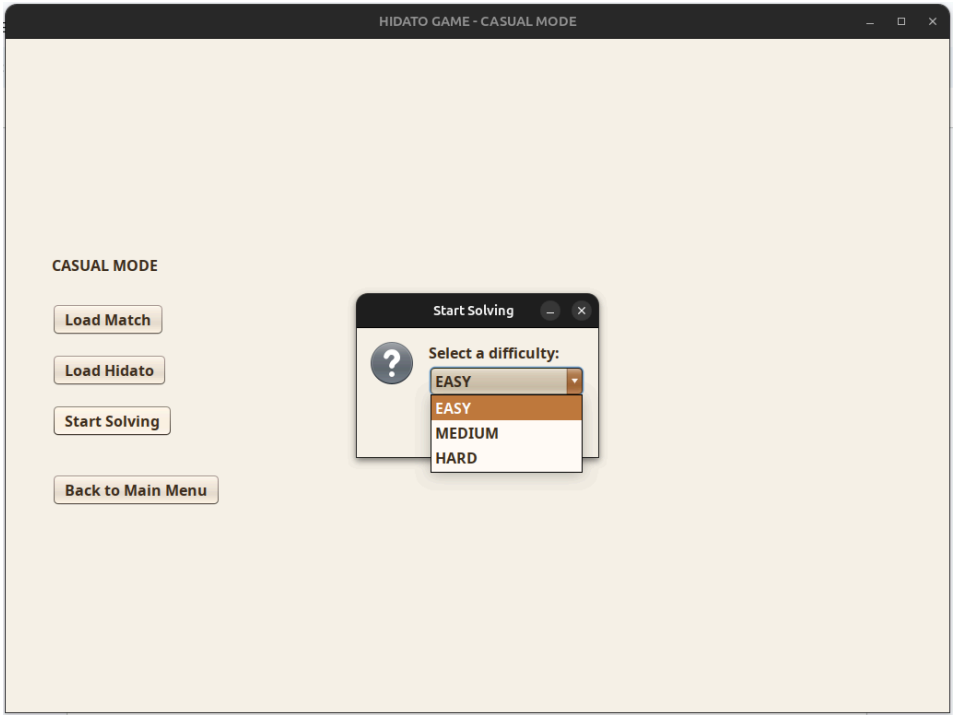


Figure 21. Difficulty selector (the same both in Casual and Ranked modes).

On the other hand, to get a hint is pretty simple. If you are currently inside a match, you must know that there is a button *Hint* and that if you press it, a random correct cell will be marked in cyan (but won't be modifiable). If the match is already solved the button is disabled. There is an edge case: you filled all cells with some errors. You therefore get the error shown in *Figure 12* and continue to solve. You are stuck and you want a hint. When pressing the button, an error pop-up will appear (*Figure 22*), as the system can not fill any cells. So first you should erase at least one cell.

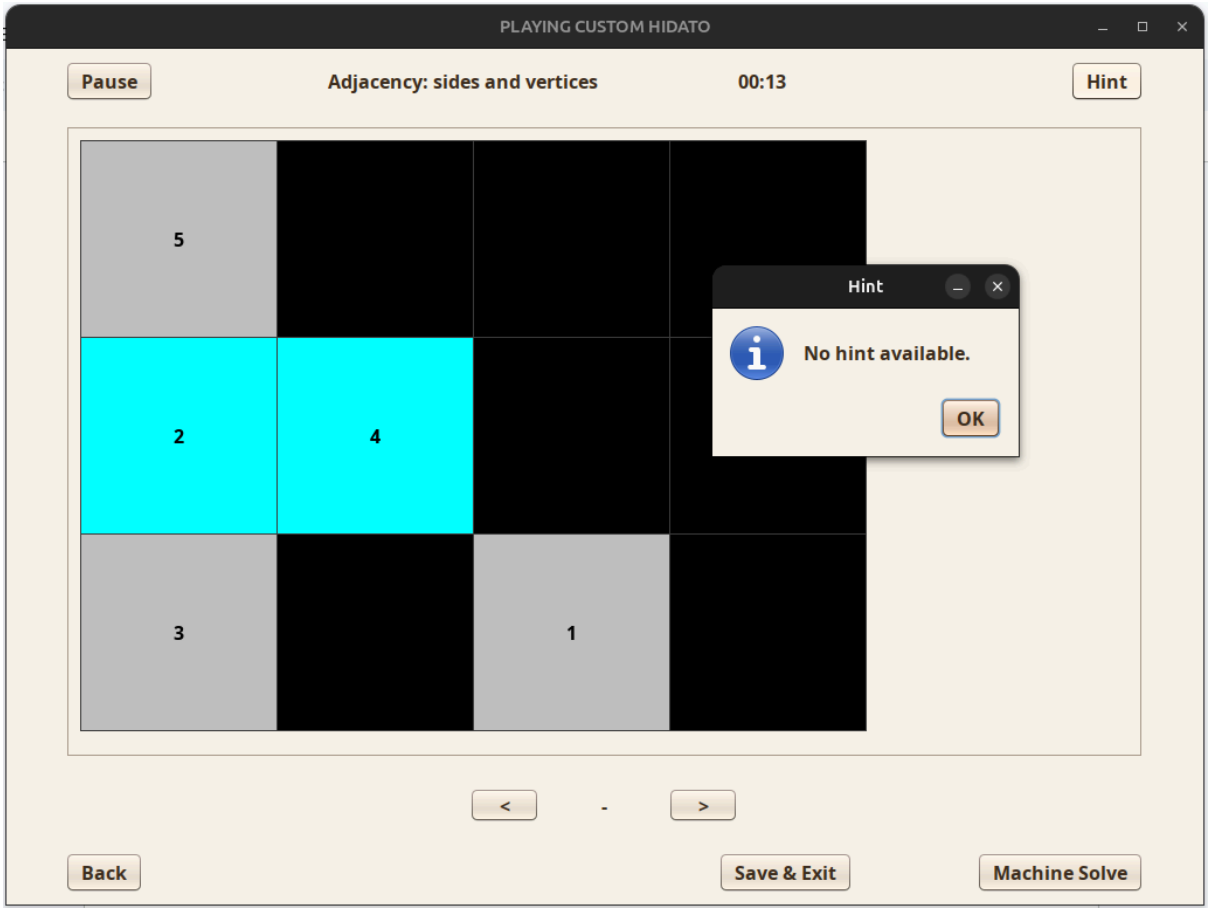


Figure 22. Getting a hint is not available as there are no cells to fill.

The application is designed to maximize ease to use. For instance, all buttons are labeled with the action they intend to execute. The main functionalities of the game are covered and detailed in this guide, but, either if you have a doubt or you want to become a Hidato Master, you can contact us through the UPC mail usernames in the front page.

Thanks for playing Hidato Game!